

Vehicle Tracking and Traffic Analysis Using Statistical Machine Vision

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1. Problem Statement

Modern cities depend on accurate traffic information to make decisions about road design, signal timing, public safety, and transportation planning. However, traffic monitoring is often performed manually by personnel reviewing video footage or counting vehicles, which is time-consuming, costly, and prone to human error.

With the widespread deployment of traffic cameras, there is a need for automated systems that can continuously detect, track, and analyze vehicle movements from video streams. Such systems can provide real-time traffic statistics, identify congestion, estimate vehicle speeds, and support data-driven transportation planning.

This project seeks to automate vehicle tracking and traffic analysis using machine vision techniques to detect and track vehicles in video sequences and generate traffic analytics.

2. Why This Problem is Important

Transportation is a critical component of economic development. Congestion causes:

- Delays in commuting
- Increased fuel consumption
- Increased transportation costs
- Reduced productivity
- Higher vehicle emissions

According to the Texas A&M Transportation Institute, traffic congestion costs billions of dollars annually in lost time and wasted fuel.

Transportation agencies require accurate traffic data to:

- Design safer roads
- Improve traffic signal timing
- Plan road expansions
- Reduce congestion
- Improve emergency response

An automated vision-based system can provide this information continuously without requiring manual observation.

3. Why the Problem is Urgent

Several factors make this problem increasingly urgent:

Growing Urbanization

The United Nations projects that nearly 70% of the world's population will live in urban areas by 2050. Increasing urban populations result in more vehicles and greater pressure on transportation infrastructure.

Increasing Number of Vehicles

Global vehicle ownership continues to rise, particularly in developing countries where road infrastructure growth often lags traffic growth.

Need for Smart Cities

Modern smart-city initiatives rely on real-time traffic monitoring systems to optimize transportation networks and improve mobility.

Road Safety

The World Health Organization reports that road traffic injuries remain a leading cause of death worldwide. Better traffic monitoring can help identify dangerous traffic patterns and support safety improvements.

4. Why Machine Vision is the Right Solution

Traffic cameras are already widely deployed at:

- Intersections
- Highways
- Parking facilities
- Toll plazas

Machine vision allows these existing cameras to become intelligent sensors without requiring additional expensive infrastructure.

Instead of human operators watching video feeds, algorithms can:

- Detect vehicles automatically
- Track vehicle movement
- Count vehicles
- Measure traffic density
- Estimate speed
- Generate reports

This enables continuous monitoring at a much lower cost.

5. Project Objectives

The primary objective is to develop a machine vision system capable of:

1. Detect a vehicle within a traffic video.
2. Track the vehicle across consecutive video frames.
3. Estimate the vehicle's speed from its tracked positions.
4. Analyze the tracking performance using correlation measures.
5. Analyze the vehicle speed signal using the Discrete Fourier Transform (DFT).
6. Visualize the tracking results and motion characteristics.

6. Relation to Statistical Machine Vision

This project integrates multiple concepts taught throughout the course:

Table 1 : Statistical Machine Vision concepts application in the project

Course Topic	Application in Project
Image Filtering	Noise reduction and smoothing
Convolution	Feature extraction
Statistical Analysis	Traffic statistics and vehicle counts
Template Matching	Vehicle detection
Video Processing	Frame-by-frame analysis
Fourier Transform	Motion frequency analysis
Object Tracking	Vehicle trajectory estimation
Peaks and Ridges	Detection of correlation maxima

These topics form the foundation of the vehicle tracking system.

Expected Outcome

The final system will automatically process traffic videos and produce:

- Vehicle trajectories
- Speed estimates
- Correlation maps
- Frequency-domain analyses

The project demonstrates how machine vision techniques can solve a real-world transportation problem while showcasing the major concepts learned throughout the Statistical Machine Vision course.

2. Data Acquisition

Traffic Video Source

The traffic video used in this project was **sample_video-2.mp4**, obtained from the [AI City Challenge](#) dataset. The video was renamed to **TrafficVideo.mp4** and used as the input for vehicle detection, tracking, and traffic analysis.

The traffic video was read using MATLAB's VideoReader. Each RGB frame was manually converted to grayscale using weighted red, green, and blue intensities and stored in a three-dimensional array. Figure 1 below shows the first, middle, and last frames of the traffic sequence used throughout the project. These images demonstrate the temporal evolution of vehicle positions and serve as the input for subsequent image processing and object tracking stages.



Figure 1 TrafficVideo.mp4 Frames

Table 2 Video Properties

Property	Value
Video format	MP4
Color format	RGB (converted to grayscale)
Number of processed frames	90

Table 3 Frame Dimensions

Property	Value
Width	1920 pixels
Height	1080 pixels
Resolution	1920 × 1080 (Full HD)

Table 4 Frame Rate

Property	Value
Frame rate	30 fps
Duration	3.0 seconds
Total frames	90

3. Methodology

The automated vehicle tracking system was implemented as a sequence of MATLAB modules, with each module performing a specific image processing task. The output of one stage serves as the input to the next, creating a complete traffic analysis pipeline.

Project Flowchart

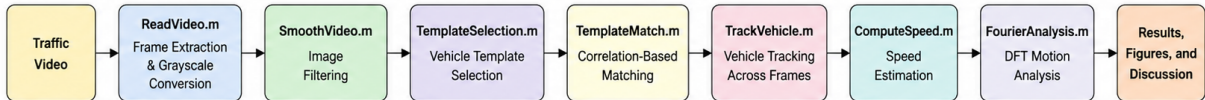


Figure 2 Project Flowchart

This workflow combines video processing, image filtering, template matching, tracking, speed estimation, and Fourier analysis into one complete machine vision system.

Table 5 MATLAB Files and Description of Purpose

Section	MATLAB File	Description
Read Video	ReadVideo.m	Reads the traffic video, extracts individual frames, converts each frame from RGB to grayscale, and stores the video as a 3-D matrix.
Smooth Frames	SmoothVideo.m	Applies a manually implemented 5×5 spatial smoothing filter to reduce image noise while preserving vehicle structures.
Select Template	TemplateSelection.m	Allows selection of a target vehicle and extracts it as the template used for matching.
Template Matching	TemplateMatch.m	Performs normalized correlation between the template and each frame to locate the vehicle and identify the correlation peak.
Vehicle Tracking	TrackVehicle.m	Repeats template matching across all frames to determine the vehicle's position and trajectory over time.
Speed Estimation	ComputeSpeed.m	Computes the vehicle's displacement between consecutive frames and estimates its speed in pixels per second.
Fourier Analysis	FourierAnalysis.m	Applies the Discrete Fourier Transform (DFT) to the estimated speed signal to analyze its frequency characteristics and dominant motion components.

4. Results

Video Smoothing

Original frame, smoothed frame, and difference image obtained after applying a manually coded 5×5 weighted smoothing kernel. The smoothing operation suppresses high-frequency variations while preserving the major structures of the vehicles and roadway. The difference image highlights the information removed by the filter, mainly edges and fine details. These results demonstrate the low-pass filtering properties of spatial convolution.



Figure 3 Video Smoothing

Template Selection

A vehicle template was selected from the middle frame of the video. The left image shows the original frame with the selected vehicle enclosed by a red rectangle, while the right image shows the extracted vehicle template. The template contains the entire vehicle and a small amount of surrounding road surface, providing a unique intensity pattern for correlation-based tracking.



Figure 4 Template Selection

Template Match

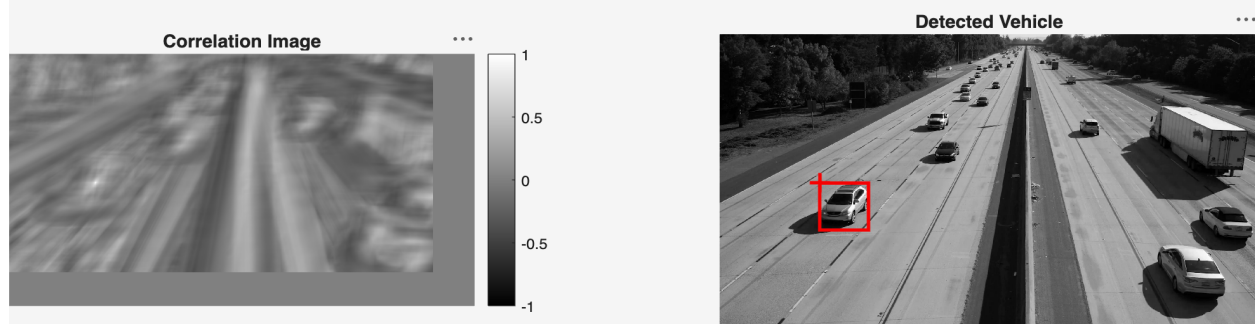


Figure 5 Template Match

Correlation image and detected vehicle location obtained using manually implemented normalized correlation. The bright peak in the correlation image corresponds to the highest similarity between the image and the selected vehicle template. The detected position is shown by the red cross and bounding box, correctly identifying the vehicle in the frame.

Peaks, Valleys, and Ridges in the Correlation Surface

The normalized correlation image was analyzed as a two-dimensional surface. The global peak (red star) corresponds to the highest correlation value and identifies the detected vehicle location, while the global valley (blue triangle) represents the lowest correlation value. The magnified view confirms that the detected vehicle is located at a local maximum. Thresholding isolates the strongest correlation response, and the three-dimensional surface visualization illustrates the overall topology of the correlation image, revealing peaks, valleys, and ridge-like structures. These results demonstrate the feature analysis techniques presented in Lecture 10 and show how extrema in a correlation surface can be used for reliable object detection.

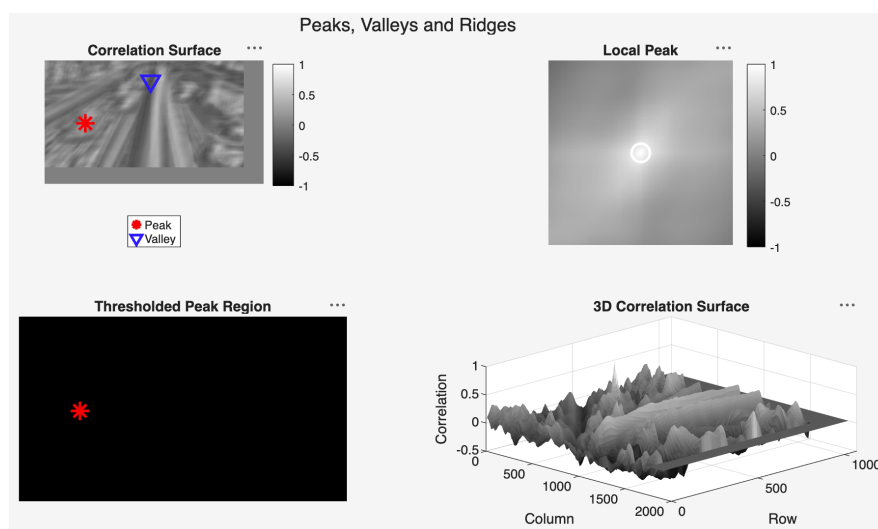


Figure 6 Peaks, Valleys, and Ridges in the Correlation Surface

Vehicle Tracking Across Frames

Vehicle tracking results for six representative frames. The red box indicates the detected vehicle location, and the red cross denotes the estimated center of the vehicle. The results demonstrate successful tracking of the selected SUV throughout the video sequence.

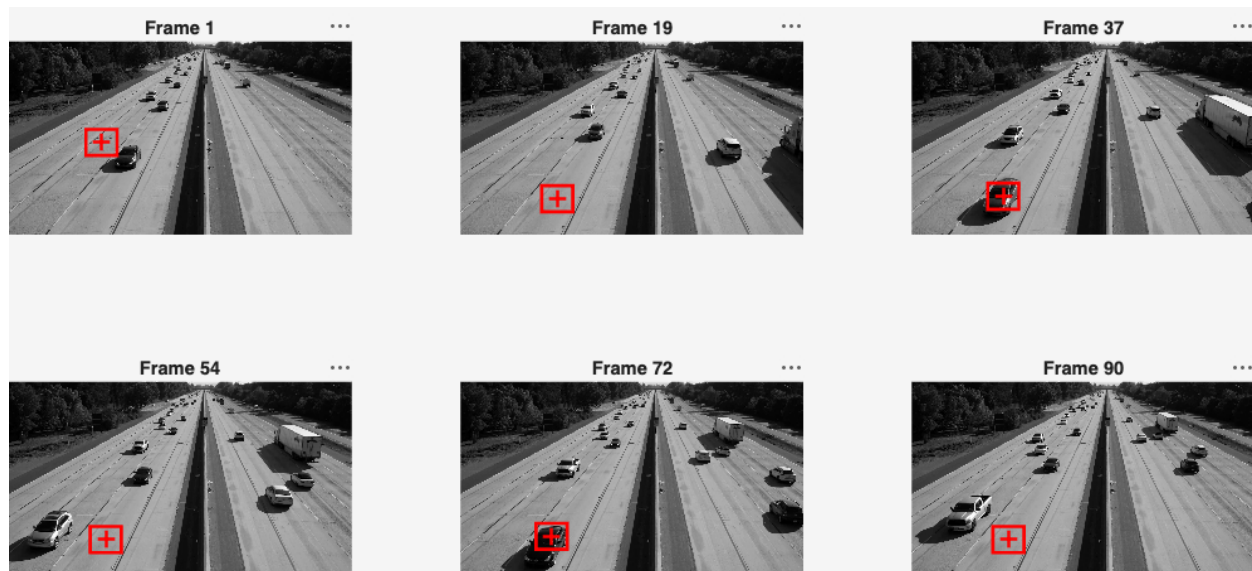


Figure 7 Vehicle Tracking Across Frames

Vehicle Trajectory Overlay

Vehicle trajectory superimposed on the traffic image. The connected red points represent the estimated vehicle centers obtained from correlation-based tracking. Although some deviations occur when the vehicle first enters the scene, the overall trajectory follows the selected SUV as it travels through the lane.

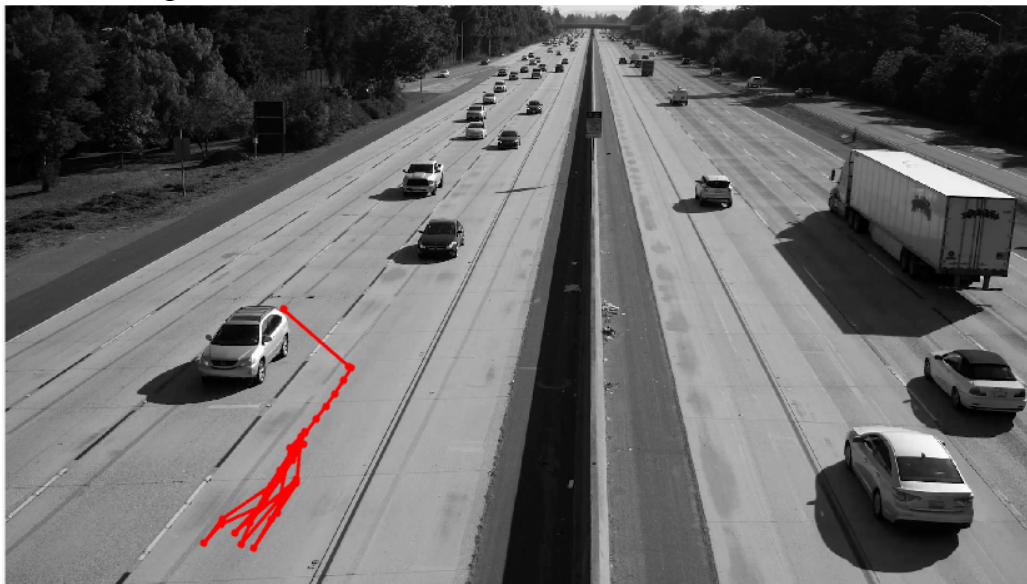


Figure 8 Vehicle Trajectory Overlay

Horizontal Position vs Frame Number

Figures 9 and 10 present the horizontal and vertical positions of the tracked vehicle as functions of frame number. For most of the video sequence, both coordinates remained nearly constant, indicating that the vehicle stayed within the same traffic lane. Several simultaneous deviations occurred in both graphs, corresponding to temporary tracking mismatches caused by similar road textures, shadows, and changes in vehicle appearance. The tracker subsequently recovered and continued following the vehicle. The agreement between the two position plots demonstrates that the tracking algorithm successfully maintained the overall vehicle trajectory despite occasional errors.

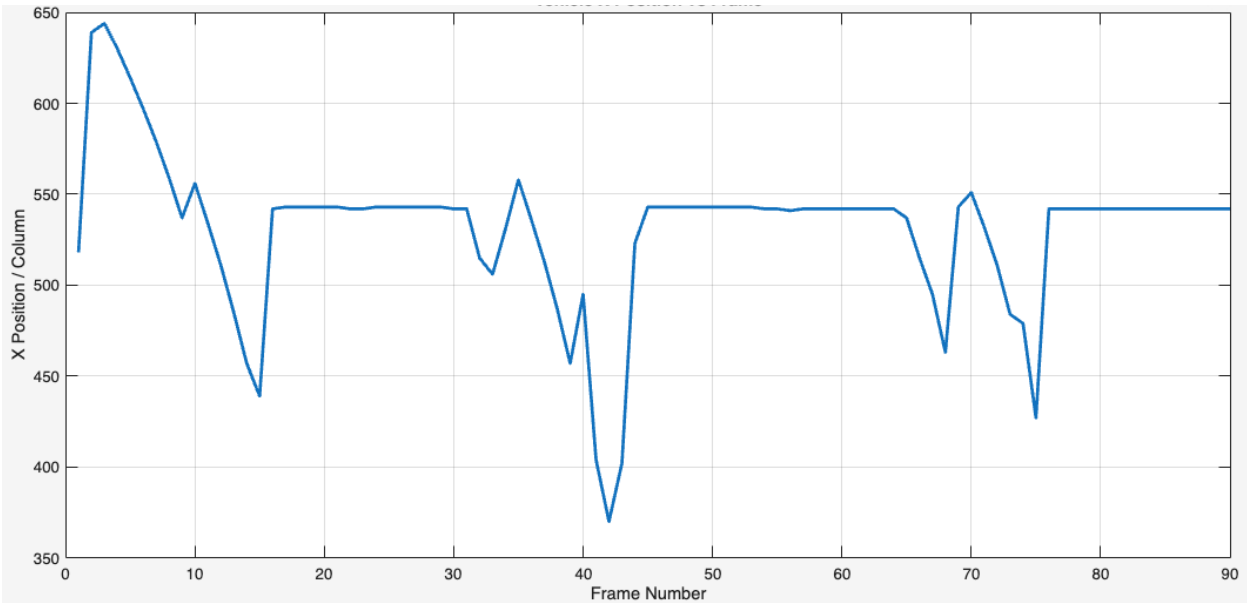


Figure 9 Horizontal(x) Position vs Frame Number

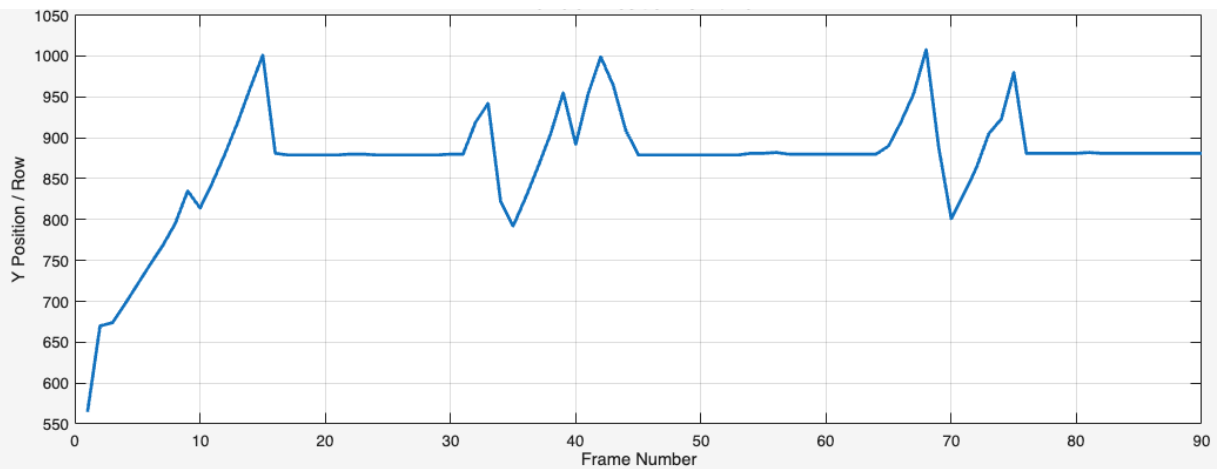


Figure 10 Vertical (y) Position vs Frame Number

Correlation Coefficient vs Frame

Figure 11 below provides a quantitative measure of tracking performance. Correlation values remained generally above 0.45 throughout the sequence, indicating that the template matching algorithm maintained a reasonable level of similarity to the target vehicle. Lower correlation values occurred during temporary tracking errors observed in Figures 9 and 10. As the tracker reacquired the vehicle, the correlation coefficient increased, demonstrating successful recovery and continued tracking.

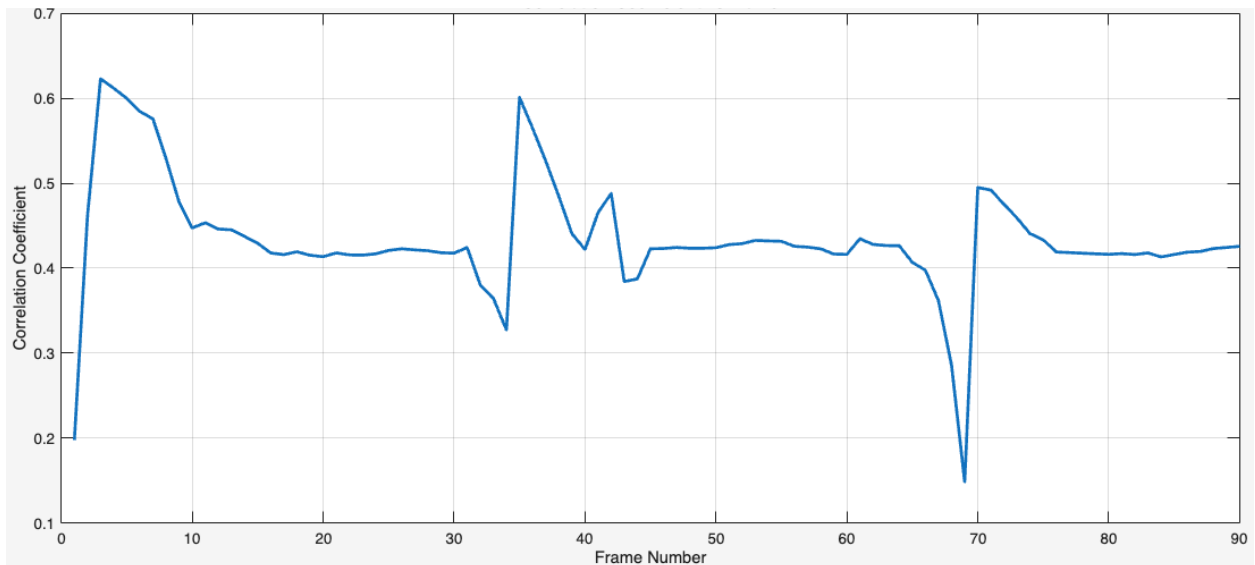


Figure 11 Correlation Coefficient vs Frame

Compute Speed

Estimated vehicle speed as a function of frame number. Speed was computed from frame-to-frame changes in the tracked vehicle center position. Spikes represent temporary tracking mismatches, while smoother regions indicate stable vehicle motion.

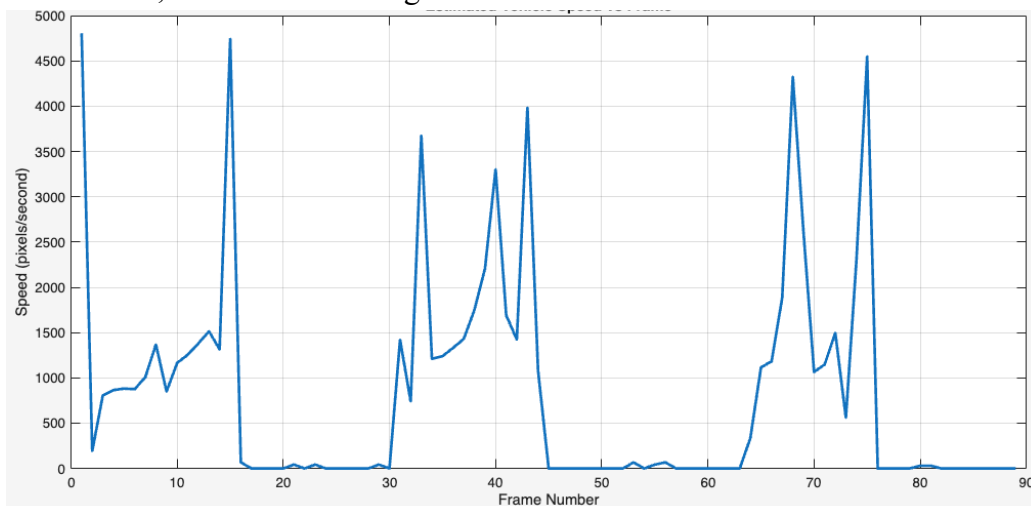


Figure 12 Estimated Vehicle Speed vs Frame

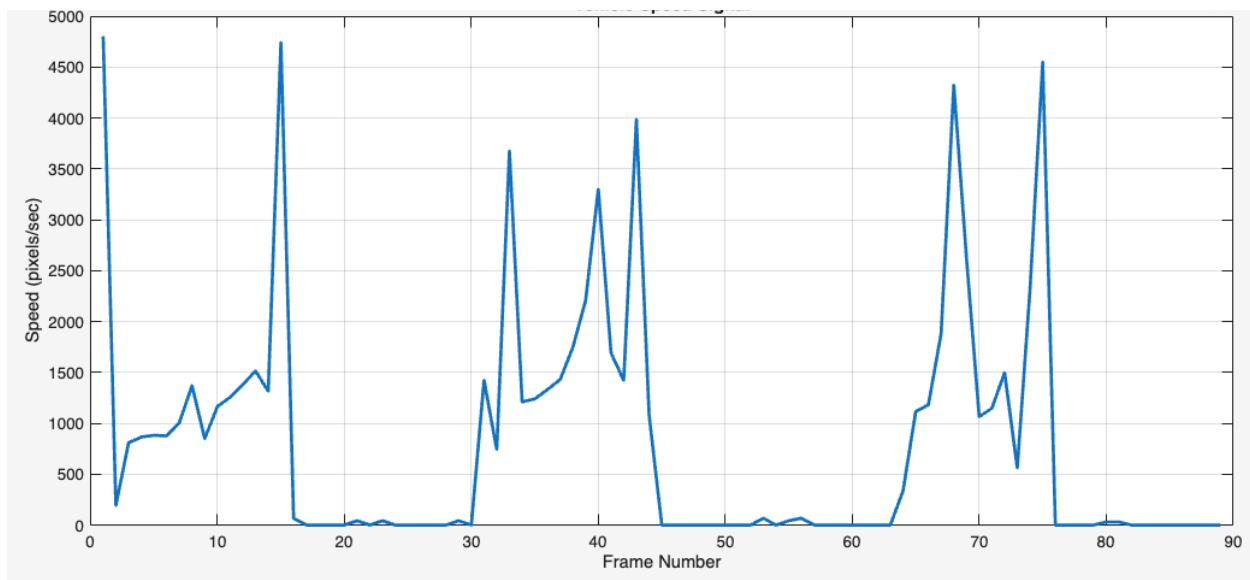


Figure 13 Vehicle Speed Signal

Figure 13 shows the estimated vehicle speed calculated from frame-to-frame changes in the tracked vehicle position. For much of the video sequence, the speed remains relatively low and stable, indicating smooth motion of the vehicle through the scene. Several large spikes are observed, particularly around frames 15, 33, 43, 68, and 75. These spikes correspond to the temporary tracking mismatches previously identified in the position plots (Figures 9 and 10), where the tracker briefly lost the vehicle and then reacquired it at a different location.

The periods where the speed approaches zero indicate intervals during which the estimated vehicle position changed very little between consecutive frames. These regions correspond to stable tracking, where the vehicle remained in approximately the same image location. The abrupt transitions between low and high speeds are therefore primarily caused by tracking errors rather than actual vehicle acceleration.

Overall, the speed signal reflects the behavior of the tracking algorithm and demonstrates how tracking inaccuracies can influence motion estimates. Despite these occasional errors, the tracker was able to repeatedly recover the target vehicle and maintain the overall motion trend throughout the video sequence.

Fourier Analysis

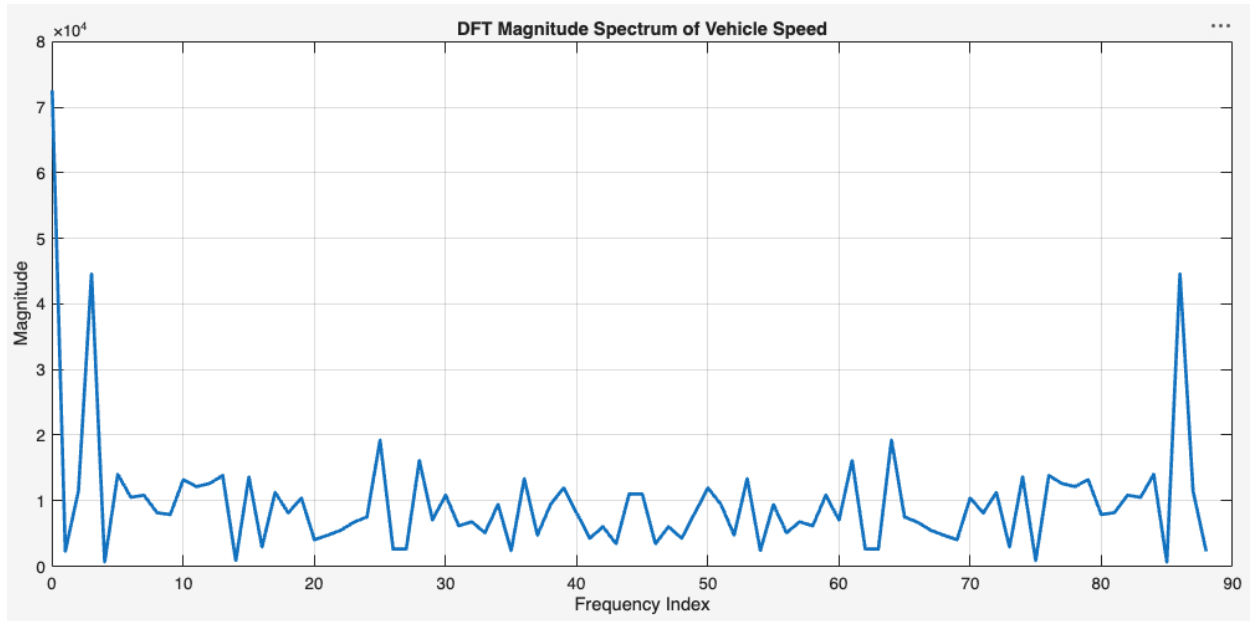


Figure 14 the Discrete Fourier Transform (DFT) Spectrum of vehicle speed

Figure 14 shows the magnitude spectrum obtained by applying the Discrete Fourier Transform (DFT) to the vehicle speed signal. A large peak is observed at the lowest frequency index, indicating that most of the energy in the signal is associated with the average vehicle motion over time. Several smaller peaks occur at higher frequencies, representing short-term variations in the estimated speed caused by acceleration, tracking noise, and temporary tracking mismatches.

The spectrum is approximately symmetric about the center of the frequency range, which is expected for the DFT of a real-valued signal. The absence of a single dominant high-frequency component suggests that the vehicle motion is relatively smooth and does not contain strong periodic oscillations. Most of the signal energy is concentrated in the low-frequency region, indicating that the overall motion changes gradually throughout the video sequence.

These results demonstrate how Fourier analysis can be used to characterize vehicle motion in the frequency domain and identify the dominant temporal components present in the tracking data.

5. Overall Discussion

The experimental results demonstrate that the manually implemented image processing pipeline successfully performs vehicle detection, tracking, speed estimation, and frequency-domain analysis. Spatial smoothing effectively reduced image noise while preserving important vehicle features for template matching. Normalized correlation accurately localized the selected vehicle throughout most of the video sequence, with temporary tracking errors occurring primarily during periods of similar background textures and illumination changes. These tracking inaccuracies propagated into the estimated speed signal as isolated spikes but did not significantly affect the overall vehicle trajectory. Finally, Fourier analysis confirmed that most motion energy was concentrated at low temporal frequencies, indicating smooth vehicle movement with only minor high-frequency fluctuations caused by tracking noise. Overall, the project demonstrates the effectiveness of classical image processing techniques for automated vehicle tracking and motion analysis without relying on machine learning methods.

6. Conclusion

The objective of this project was to develop an image processing system for detecting, tracking, and analyzing the motion of a vehicle in a traffic video. This objective was successfully achieved using MATLAB through spatial smoothing, normalized correlation, vehicle tracking, speed estimation, and Discrete Fourier Transform (DFT) analysis.

The smoothing filter reduced image noise while preserving important vehicle features for reliable template matching. The tracking algorithm successfully followed the selected vehicle throughout most of the video, generating its trajectory and speed estimates. Although temporary tracking mismatches produced occasional errors, the tracker consistently recovered the target and maintained the overall motion trend. Fourier analysis showed that most of the motion energy was concentrated at low frequencies, indicating relatively smooth vehicle movement.

Overall, the project successfully demonstrated the complete workflow of vehicle detection, tracking, speed estimation, and frequency-domain analysis using classical image processing techniques, providing a solid foundation for more advanced traffic monitoring and computer vision applications.